Mathematical Derivations of Fast Topological Data Analysis (Ellipse Fitting via Takens Embedding)

Contents

1 TDA Time Delay and Window Duration

Goal: Translate $F_{\max}$ and $F_{\min}$ into the two control parameters of the Takens embedding: the time delay τ and the window duration T_w. Applying the Razmarashooli method, which sets τ to one quarter of the dominant period and T_w to one full period of the lowest frequency.

Following the Razmarashooli method, the time delay τ and window duration T_w are defined as:

$$\tau = \frac{0.25}{F_{\max}} \quad \text{(seconds)} \tag{1}$$

$$\tau_s = \text{round}\left(\frac{\tau}{\Delta t}\right) \quad \text{(samples)} \tag{2}$$

$$T_w = \frac{1}{F_{\min}} \quad \text{(seconds)} \tag{3}$$

$$N_w = \text{round}\left(\frac{T_w}{\Delta t}\right) \quad \text{(samples per window)} \tag{4}$$

Rationale for $\tau = T/4$: For a sinusoid $x(t) = A\sin(2\pi F_{\max} t)$, a quarter-period delay produces a cosine:

$$x(t+\tau) = A\sin\left(2\pi F_{\max} t + \frac{\pi}{2}\right) = A\cos(2\pi F_{\max} t) \tag{5}$$

Plotting $x(t)$ vs. $x(t+\tau)$ traces a perfect circle, maximising the 2-D spread of the point cloud and giving the best-conditioned ellipse fit. Any other delay angle produces an ellipse, but $\tau = T/4$ is the unique delay that makes the two embedded dimensions statistically independent for a sinusoid.

2 Zero-Padding

Goal: Ensure the sliding window is valid from sample index $i = 0$ without accessing samples that do not exist in the original signal. Prepending N_{pad} zeros to the signal before the main loop begins.

The earliest sample accessed by the window at position $i = 0$ is at conceptual index:

$$\ell = -(N_w - 1) - \tau_s < 0 \tag{6}$$

Since $\ell < 0$, samples before the signal start are required. The required number of prepended zeros is:

$$N_{\text{pad}} = |\ell| = N_w - 1 + \tau_s \tag{7}$$

The zero-padded signal is defined as:

$$x_{\text{pad}}(n) = \begin{cases} 0 & n = 0, \ldots, N_{\text{pad}} - 1 \quad \text{(prepended zeros)} \\ x(n - N_{\text{pad}}) & n = N_{\text{pad}}, \ldots, N_{\text{pad}} + N - 1 \quad \text{(original signal)} \end{cases} \tag{8}$$

3 Takens Delay Embedding

Goal: Convert the 1-D signal within each window into a 2-D point cloud that traces an ellipse in phase space. Pairing each sample with a delayed copy of itself, separated by τ_s samples. For a periodic signal this produces a closed elliptical loop; deviations from the loop reveal dynamic changes.

For window starting at sample index i, the two-dimensional point cloud $\mathbf{P}$ is constructed as:

$$\mathbf{P} = \begin{bmatrix} x(i) & x(i+\tau_s) \\ x(i+1) & x(i+\tau_s+1) \\ \vdots & \vdots \\ x(i+N_w-\tau_s-1) & x(i+N_w-1) \end{bmatrix} \in \mathbb{R}^{(N_w-\tau_s)\times 2} \tag{9}$$

Each row k is a 2-D point in the reconstructed phase space:

$$\mathbf{p}_k = \big(x(i+k-1),\ x(i+k-1+\tau_s)\big), \qquad k = 1, \ldots, N_w - \tau_s \tag{10}$$

Let $n = N_w - \tau_s$ denote the number of points per window.

4 General Conic Equation and Ellipse Constraint

Goal: Define the mathematical form of the curve we are fitting (an ellipse), and establish the algebraic constraint that guarantees the fitted curve is always an ellipse — never a hyperbola or parabola. Useing the general degree-2 implicit equation and apply the Fitzgibbon constraint $4ac - b^2 = 1$.

4.1 General Conic

Any conic section in $\mathbb{R}^2$ satisfies the implicit equation:

$$\boxed{F(x, y) = ax^2 + bxy + cy^2 + dx + ey + f = 0} \tag{11}$$

with coefficient vector $\mathbf{a} = [a,\ b,\ c,\ d,\ e,\ f]^\top$. The coefficients a, b, c control shape and tilt; d, e, f control position and offset.

4.2 Ellipse Discriminant Condition

The conic type is determined by the discriminant $\Delta = b^2 - 4ac$:

$$\Delta = \begin{cases} < 0 & \Rightarrow \text{ellipse} \\ = 0 & \Rightarrow \text{parabola} \\ > 0 & \Rightarrow \text{hyperbola} \end{cases} \tag{12}$$

Without a constraint, an unconstrained fit might return a hyperbola or parabola that also passes through the data. We must explicitly enforce $\Delta < 0$.

4.3 Fitzgibbon Algebraic Constraint

To guarantee an ellipse solution, Fitzgibbon et al. enforce:

$$\mathbf{a}^\top \mathbf{C}\, \mathbf{a} = 1, \qquad \text{where} \quad 4ac - b^2 = 1 \tag{13}$$

This simultaneously fixes the scale ambiguity of $\mathbf{a}$ (since any scalar multiple gives the same curve) and guarantees the discriminant condition $\Delta < 0$. The full 6×6 constraint matrix $\mathbf{C}$ is constructed so that $\mathbf{a}^\top \mathbf{C}\mathbf{a}$ evaluates exactly to $4ac - b^2$:

$$\mathbf{C} = \begin{bmatrix} 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{14}$$

The lower 3×3 block is zero because d, e, f do not appear in the discriminant condition.

5 Design Matrix and Scatter Matrix

Goal: Organise all n data points into a compact matrix form that the optimisation algorithm can work with. . Each point (x_k, y_k) contributes one row to the design matrix $\mathbf{D}$ by evaluating the six conic monomials. Multiplying $\mathbf{D}^\top\mathbf{D}$ accumulates information from all n points into the fixed-size 6×6 scatter matrix $\mathbf{S}$.

5.1 Design Matrix

Each point (x_k, y_k) contributes one row:

$$\mathbf{d}_k = \left[x_k^2,\ x_k y_k,\ y_k^2,\ x_k,\ y_k,\ 1\right] \in \mathbb{R}^{1\times 6} \tag{15}$$

The full design matrix stacking all n points is:

$$\mathbf{D} = \begin{bmatrix} x_1^2 & x_1y_1 & y_1^2 & x_1 & y_1 & 1 \\ x_2^2 & x_2y_2 & y_2^2 & x_2 & y_2 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_n^2 & x_ny_n & y_n^2 & x_n & y_n & 1 \end{bmatrix} \in \mathbb{R}^{n\times 6} \tag{16}$$

$\mathbf{D}$ is partitioned by column type (quadratic vs. linear monomials):

$$\mathbf{D} = [\,\mathbf{D}_1 \mid \mathbf{D}_2\,], \qquad \mathbf{D}_1 = [x^2,\, xy,\, y^2], \quad \mathbf{D}_2 = [x,\, y,\, 1] \tag{17}$$

with corresponding coefficient partition $\mathbf{a}_1 = [a, b, c]^\top$ and $\mathbf{a}_2 = [d, e, f]^\top$.

5.2 Scatter Matrix

The scatter matrix accumulates all n points into a 6×6 summary via:

$$\mathbf{S} = \mathbf{D}^\top \mathbf{D} \in \mathbb{R}^{6\times 6}, \qquad S_{ij} = \sum_{k=1}^{n} D_{ki}\, D_{kj} \tag{18}$$

Every entry is a sum over all n points. For example $S_{11} = \sum x_k^4$ and $S_{66} = n$. No information is lost: $\mathbf{S}$ is a complete statistical summary of the point cloud. Partitioned into 3×3 blocks:

$$\mathbf{S} = \begin{bmatrix} \mathbf{S}_{11} & \mathbf{S}_{12} \\ \mathbf{S}_{21} & \mathbf{S}_{22} \end{bmatrix}, \qquad \mathbf{S}_{21} = \mathbf{S}_{12}^\top \tag{19}$$

where explicitly:

$$\mathbf{S}_{11} = \begin{bmatrix} \sum x^4 & \sum x^3y & \sum x^2y^2 \\ \sum x^3y & \sum x^2y^2 & \sum xy^3 \\ \sum x^2y^2 & \sum xy^3 & \sum y^4 \end{bmatrix}, \quad \mathbf{S}_{22} = \begin{bmatrix} \sum x^2 & \sum xy & \sum x \\ \sum xy & \sum y^2 & \sum y \\ \sum x & \sum y & n \end{bmatrix} \tag{20}$$

6 Constrained Least-Squares Problem

Goal: Find the coefficient vector $\mathbf{a}$ that makes the ellipse pass as close as possible to all n data points while remaining a valid ellipse. Minimizing the total squared algebraic distance $\mathbf{a}^\top\mathbf{S}\mathbf{a}$ subject to the ellipse constraint, using a Lagrange multiplier. This converts the constrained optimisation into a generalised eigenvalue problem.

6.1 Lagrangian Formulation

We minimise the sum of squared algebraic distances subject to the ellipse constraint (13):

$$\min_{\mathbf{a}}\ \mathbf{a}^\top \mathbf{S}\,\mathbf{a} \quad \text{subject to} \quad \mathbf{a}^\top \mathbf{C}\,\mathbf{a} = 1 \tag{21}$$

Introducing Lagrange multiplier λ and differentiating:

$$\mathcal{L}(\mathbf{a}, \lambda) = \mathbf{a}^\top \mathbf{S}\,\mathbf{a} - \lambda\big(\mathbf{a}^\top \mathbf{C}\,\mathbf{a} - 1\big) \tag{22}$$

Setting $\partial\mathcal{L}/\partial\mathbf{a} = \mathbf{0}$ gives the **generalised eigenvalue problem**:

$$\boxed{\mathbf{S}\,\mathbf{a} = \lambda\,\mathbf{C}\,\mathbf{a}} \tag{23}$$

Pre-multiplying by $\mathbf{a}^\top$ shows that $\lambda = \mathbf{a}^\top\mathbf{S}\mathbf{a}$, i.e. λ equals the objective (fitting error). The smallest $|\lambda|$ gives the best fit.

7 Schur Complement Reduction ($6 \times 6 \to 3 \times 3$)

Goal: Reduce the expensive 6×6 generalised eigenvalue problem to a cheap 3×3 standard one. Exploiting the block structure of $\mathbf{C}$ (its lower 3×3 block is zero) to eliminate $\mathbf{a}_2$ algebraically from the system, leaving only a 3×3 problem in $\mathbf{a}_1$.

7.1 Block System

Writing (23) in block form:

$$\begin{bmatrix} \mathbf{S}_{11} & \mathbf{S}_{12} \\ \mathbf{S}_{21} & \mathbf{S}_{22} \end{bmatrix} \begin{bmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \end{bmatrix} = \lambda \begin{bmatrix} \mathbf{C}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{a}_1 \\ \mathbf{a}_2 \end{bmatrix} \tag{24}$$

where $\mathbf{C}_1$ is the top-left 3×3 block:

$$\mathbf{C}_1 = \begin{bmatrix} 0 & 0 & 2 \\ 0 & -1 & 0 \\ 2 & 0 & 0 \end{bmatrix} \tag{25}$$

This gives two block equations:

$$\mathbf{S}_{11}\,\mathbf{a}_1 + \mathbf{S}_{12}\,\mathbf{a}_2 = \lambda\,\mathbf{C}_1\,\mathbf{a}_1 \tag{I}$$

$$\mathbf{S}_{21}\,\mathbf{a}_1 + \mathbf{S}_{22}\,\mathbf{a}_2 = \mathbf{0} \tag{II}$$

7.2 Elimination of $\mathbf{a}_2$

Equation (II) contains no λ (because the lower block of $\mathbf{C}$ is zero), so it can be solved directly for $\mathbf{a}_2$:

$$\mathbf{a}_2 = -\mathbf{S}_{22}^{-1}\,\mathbf{S}_{12}^\top\,\mathbf{a}_1 \tag{26}$$

This is the back-substitution formula: once $\mathbf{a}_1$ is known, $\mathbf{a}_2$ is completely determined.

7.3 Schur Complement

Substituting (26) into (I) and factoring out $\mathbf{a}_1$ yields:

$$\boxed{\mathbf{S}^* = \mathbf{S}_{11} - \mathbf{S}_{12}\,\mathbf{S}_{22}^{-1}\,\mathbf{S}_{12}^\top} \tag{27}$$

$\mathbf{S}^*$ is the **Schur complement**: the part of the quadratic information in $\mathbf{S}_{11}$ that cannot be explained by the linear terms. The 6×6 problem reduces to a 3×3 generalised problem:

$$\mathbf{S}^*\,\mathbf{a}_1 = \lambda\,\mathbf{C}_1\,\mathbf{a}_1 \tag{28}$$

Pre-multiplying by $\mathbf{C}_1^{-1}$ converts to standard form:

$$\underbrace{\mathbf{C}_1^{-1}\,\mathbf{S}^*}_{\mathbf{A}}\,\mathbf{a}_1 = \lambda\,\mathbf{a}_1 \tag{29}$$

8 Analytical Inverse of $\mathbf{S}_{22}$

Goal: Compute $\mathbf{S}_{22}^{-1}$ needed for both the Schur complement and the back-substitution. Since $\mathbf{S}_{22}$ is always 3×3, its inverse is computed in closed form using the cofactor method — no iterative algorithm needed. A small regularisation $\varepsilon = 10^{-6}$ is added to the diagonal first to prevent singularity when the data is nearly collinear.

With regularisation $\varepsilon = 10^{-6}$ added to the diagonal, let:

$$\mathbf{S}_{22} + \varepsilon \mathbf{I} = \begin{bmatrix} \alpha & \beta & \gamma \\ \beta & \delta & \zeta \\ \gamma & \zeta & \eta \end{bmatrix} \tag{30}$$

The determinant (required for all cofactor entries) is:

$$\det = \alpha(\delta\eta - \zeta^2) - \beta(\beta\eta - \zeta\gamma) + \gamma(\beta\zeta - \delta\gamma) \tag{31}$$

The inverse entries (exploiting symmetry $Q_{ij} = Q_{ji}$) are:

$$Q_{11} = \frac{\delta\eta - \zeta^2}{\det} \qquad Q_{12} = -\frac{\beta\eta - \gamma\zeta}{\det} \qquad Q_{13} = \frac{\beta\zeta - \gamma\delta}{\det} \tag{32}$$

$$Q_{22} = \frac{\alpha\eta - \gamma^2}{\det} \qquad Q_{23} = -\frac{\alpha\zeta - \gamma\beta}{\det} \qquad Q_{33} = \frac{\alpha\delta - \beta^2}{\det} \tag{33}$$

9 Matrix A and Eigenvalue Problem

Goal: Form the 3×3 matrix $\mathbf{A}$ and solve its eigenvalue problem to find the quadratic coefficients $\mathbf{a}_1 = [a, b, c]^\top$. Since $\mathbf{C}_1$ is a fixed constant matrix, its inverse $\mathbf{C}_1^{-1}$ is precomputed analytically and hardcoded. The eigenvector with the smallest $|\lambda|$ is selected because λ equals the fitting error.

9.1 Computing A

$\mathbf{A} = \mathbf{C}_1^{-1}\,\mathbf{S}^*$, where $\mathbf{C}_1^{-1}$ is a fixed constant hardcoded in the code (it never changes because $\mathbf{C}_1$ depends only on the mathematical ellipse constraint, not on any data):

$$\det(\mathbf{C}_1) = 4, \qquad \mathbf{C}_1^{-1} = \begin{bmatrix} 0 & 0 & 0.5 \\ 0 & -1 & 0 \\ 0.5 & 0 & 0 \end{bmatrix} \tag{34}$$

9.2 Standard Eigenvalue Problem

The 3×3 standard eigenvalue problem is solved:

$$\mathbf{A}\,\mathbf{a}_1 = \lambda\,\mathbf{a}_1 \tag{35}$$

yielding eigenvalue-eigenvector pairs $\{(\lambda_1, \mathbf{v}_1), (\lambda_2, \mathbf{v}_2), (\lambda_3, \mathbf{v}_3)\}$. The optimal quadratic coefficients are the eigenvector with the smallest absolute eigenvalue (smallest fitting error):

$$\mathbf{a}_1 = \mathbf{v}_{k^*}, \qquad k^* = \arg\min_k |\lambda_k| \tag{36}$$

10 Back-Substitution for Linear Coefficients

Goal: Recover the linear coefficients $[d, e, f]^\top$ from the quadratic ones $[a, b, c]^\top$. Substituting $\mathbf{a}_1$ into the exact algebraic formula (26) derived in Section 6.2. This is not an approximation — it is the unique linear-coefficient vector that minimises the fitting error given $\mathbf{a}_1$.

Once $\mathbf{a}_1 = [a, b, c]^\top$ is known, the linear coefficients $\mathbf{a}_2 = [d, e, f]^\top$ follow exactly:

$$\mathbf{a}_2 = -\mathbf{S}_{22}^{-1}\,\mathbf{S}_{12}^\top\,\mathbf{a}_1 \tag{37}$$

The full six-parameter vector is assembled as:

$$\mathbf{a} = \begin{bmatrix}\mathbf{a}_1\\ \mathbf{a}_2\end{bmatrix} = [a,\ b,\ c,\ d,\ e,\ f]^\top \tag{38}$$

11 Fitzgibbon Normalisation

Goal: Fix the scale ambiguity of the eigenvector (which is only defined up to a scalar multiple) so that coefficients from different windows are comparable. Dividing $\mathbf{a}$ by $\sqrt{|4ac - b^2|}$, which forces the Fitzgibbon constraint to equal exactly 1. This does not change the ellipse shape — only its numerical scale.

The normalisation scale factor is:

$$\sigma = \sqrt{\left|\,\mathbf{a}^\top\mathbf{C}\,\mathbf{a}\,\right|} = \sqrt{|4ac - b^2|} \tag{39}$$

$$\mathbf{a}_{\text{norm}} = \frac{\mathbf{a}}{\sigma} \tag{40}$$

Verification — the constraint is satisfied after normalisation:

$$\mathbf{a}_{\text{norm}}^\top\,\mathbf{C}\,\mathbf{a}_{\text{norm}} = \frac{1}{\sigma^2}\,\mathbf{a}^\top\mathbf{C}\,\mathbf{a} = \frac{4ac - b^2}{|4ac - b^2|} = 1 \quad ✓ \tag{41}$$

12 Conic to Parametric Conversion

Goal: Convert the six algebraic coefficients $[a, b, c, d, e, f]$ into human-interpretable geometric parameters: ellipse centre (x_0, y_0), semi-axes (r_1, r_2), and tilt angle θ. Working through five sub-steps — find the critical point of F (centre), evaluate F there, diagonalise the quadratic part (angle), compute the eigenvalues of the quadratic form (axis scaling), and extract the semi-axes.

12.1 Step 1: Ellipse Centre

The centre is the unique point where both partial derivatives of $F(x, y)$ vanish simultaneously ($\partial F/\partial x = \partial F/\partial y = 0$), solved by Cramer's rule with denominator $\Delta = b^2 - 4ac < 0$:

$$\begin{bmatrix}2a & b\\ b & 2c\end{bmatrix}\begin{bmatrix}x_0\\ y_0\end{bmatrix} = \begin{bmatrix}-d\\ -e\end{bmatrix} \implies x_0 = \frac{2cd - be}{b^2 - 4ac}, \qquad y_0 = \frac{2ae - bd}{b^2 - 4ac} \tag{42}$$

12.2 Step 2: Value at Centre

Evaluating F at the centre gives the scalar a', which acts as a normalisation constant for the semi-axes. For a valid (non-degenerate) ellipse, $a' < 0$:

$$a' = ax_0^2 + bx_0y_0 + cy_0^2 + dx_0 + ey_0 + f \tag{43}$$

12.3 Step 3: Tilt Angle

The tilt angle θ is the rotation that eliminates the cross-term bxy by diagonalising the 2×2 quadratic form matrix $\mathbf{Q} = \left[\begin{smallmatrix} a & b/2 \\ b/2 & c \end{smallmatrix}\right]$:

$$\tan(2\theta) = \frac{b}{a-c} \implies \theta = \begin{cases} 0 & \text{if } b \approx 0 \text{ (already axis-aligned)} \\ \pi/4 & \text{if } a \approx c \text{ (isotropic quadratic part)} \\ \frac{1}{2}\arctan\left(\frac{b}{a-c}\right) & \text{otherwise} \end{cases} \tag{44}$$

12.4 Step 4: Eigenvalues of the Quadratic Form

The eigenvalues of $\mathbf{Q}$ determine the curvature of the ellipse in each axis direction. They are found from the characteristic equation of the 2×2 matrix $\mathbf{Q}$:

$$\lambda_{1,2} = \frac{(a+c) \pm \sqrt{(a-c)^2 + b^2}}{2} \tag{45}$$

12.5 Step 5: Semi-Axes

After translating to the centre and rotating by θ, the conic reduces to the canonical form $\lambda_1 u^2 + \lambda_2 v^2 = -a'$. Comparing with the standard ellipse form $u^2/r_1^2 + v^2/r_2^2 = 1$ gives the semi-axes:

$$r_1 = \sqrt{\frac{-a'}{\lambda_1}}, \qquad r_2 = \sqrt{\frac{-a'}{\lambda_2}} \tag{46}$$

with $r_{\text{major}} = \max(r_1, r_2)$ and $r_{\text{minor}} = \min(r_1, r_2)$.

12.6 Parametric Form

The complete fitted ellipse in the original coordinate system:

$$\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} + \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} r_1 \cos t \\ r_2 \sin t \end{bmatrix}, \qquad t \in [0,\, 2\pi) \tag{47}$$

13 Min-Max Normalisation for Stacked Visualisation

Goal: Display all five ellipse parameters (a, b, c, d, e) on the same plot with equal visual amplitude so they can be compared directly. Normalising the Min-max of each parameter to $[0, 1]$ independently, then add a fixed vertical offset to separate the five bands. This guarantees equal band height regardless of the raw magnitude of each parameter.

For each parameter $j \in \{1, \ldots, 5\}$:

$$v_{\text{norm},i}^{(j)} = \frac{v_i^{(j)} - \min_i v_i^{(j)}}{\max_i v_i^{(j)} - \min_i v_i^{(j)}} \in [0,\, 1] \tag{48}$$

The stacked (offset) display value is:

$$\tilde{v}_i^{(j)} = v_{\text{norm},i}^{(j)} + \delta_j, \qquad \delta_j = (5-j) \times \Delta_{\text{sep}} \tag{49}$$

where $\Delta_{\text{sep}} = 1.15$ is the vertical separation, placing parameter a ($j = 1$) at the top (offset $= 4 \times 1.15$) and e ($j = 5$) at the bottom (offset $= 0$). Y-axis tick labels are placed at the centre of each band:

$$y_{\text{tick}}^{(j)} = \delta_j + 0.5 \tag{50}$$